

8 Programming

Cambridge IGCSE Computer Science (2210) – revision notes

In this chapter: variables/constants/data types · input & output · sequence, selection, iteration · totalling/counting/string handling · operators · procedures & functions · library routines · maintainable programs · arrays (1D & 2D) · file handling.

8.1.1–8.1.2 Variables, Constants & Data Types

Variable: a named data store whose value CAN change while the program runs.

Constant: a named data store whose value CANNOT change while the program runs (often shown in CAPITALS or starting "Const").

Declaring = stating a variable/constant's name (and usually its data type) before use. Python does NOT require declarations; Visual Basic and Java DO.

Data type	Description
integer	positive/negative whole number
real	positive/negative number with a fractional part
char	a single character
string	zero or more characters (can be empty)
Boolean	only TRUE or FALSE

Common mistake: a number stored as a STRING cannot be used in calculations — it must be cast/converted first (e.g. Python's `int()` / `float()`).

8.1.3 Input & Output

Every INPUT should have a prompt telling the user what to enter; every OUTPUT should have a message explaining the result. All keyboard input defaults to a string, so numeric input must be cast to the right type.

	<i>Python</i>	<i>Visual Basic</i>	<i>Java</i>
Input	<pre>x = float(input("Enter x "))</pre>	<pre>x = Decimal.Parse(Console.ReadLine())</pre>	<pre>x = myObj.nextDouble()</pre>
Output	uses comma ,	uses &	uses +

8.1.4 Core Programming Concepts

(a) Sequence

The ORDER of steps matters – wrong order gives wrong results. Classic exam trap: reading a value AFTER using it in a total, or testing for a sentinel value in the wrong place (see worked example in textbook: moving INPUT before/after the totalling step changes the answer completely).

(b) Selection – IF and CASE

Pseudocode	Python	Visual Basic	Java
IF...THEN...ELSE...ENDIF	<code>if: / else: (colon+indent)</code>	<code>If...Then...Else...End If</code>	<code>if(){...} else{...}</code>
CASE OF...OTHERWISE...ENDCASE	<code>elif chain</code>	<code>Select Case...Case Else...End Select</code>	<code>switch{case: ... break; default:}</code>

Common mistake: forgetting `break;` in a Java switch – execution "falls through" into the next case!

(c) Iteration

Loop type	Python	Visual Basic	Java
Count-controlled (FOR)	<code>for i in range(1,11,2):</code>	<code>For i=1 To 10 Step 2 ... Next</code>	<code>for(int i=1;i<=10;i+=2){}</code>
Pre-condition (WHILE) – may run 0 times	<code>while cond:</code>	<code>While cond ... End While</code>	<code>while(cond){}</code>
			<code>do{}while(cond);</code>

Post-condition (REPEAT) – runs ≥1 time	Python has none (use while True + break)	Do ... Loop Until cond	
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Careful: Visual Basic's `Loop Until` exits when TRUE, but Java's `while()` in a do-loop keeps going WHILE true (opposite condition sense) – same logic, different wording!

(d) Totalling & counting

```
TotalWeight = TotalWeight + Weight    # totalling – same in all 3
languages                               languages
NumberOfItems = NumberOfItems + 1      # counting
NumberOfItems += 1                     # Python/VB shorthand
NumberOfItems++;                       # Java shorthand
```

(e) String handling

Strings are indexed from position 0 in Python/VB/Java, but position 1 in this syllabus's pseudocode.

Task	Pseudocode	Python	Java/VB
Length	LENGTH(Str)	len(Str)	Str.length() / Str.Length()
Substring	SUBSTRING(Str, start, len)	Str[a:b]	Str.substring(a, b)
Upper	UCASE(Str)	Str.upper()	Str.toUpperCase()
Lower	LCASE(Str)	Str.lower()	Str.toLowerCase()

(f) Operators

Type	Operators
Arithmetic	+ − * / ^ (power) · MOD (remainder) · DIV (integer/quotient division)
Logical (comparison)	> < = >= <= <> (not equal)
Boolean	AND (both true) · OR (either true) · NOT (negation)

MOD & DIV: $17 \text{ MOD } 5 = 2$ (remainder) · $17 \text{ DIV } 5 = 3$ (whole number quotient). Python: % and //. Visual Basic: Mod and \.

8.1.5 Nested Statements

Constructs can be nested inside each other — e.g. a FOR loop inside a FOR loop inside a FOR loop (three levels: class → subject → test). Work from the INNERMOST loop outwards when designing or tracing.

8.1.6 Procedures & Functions

Procedure: a named group of statements, called to perform a task — does NOT return a value. (Python: "void function"; VB: "Sub"; Java: "method".)

Function: a named group of statements that ALWAYS returns a value using the `RETURN` keyword. Can be used directly inside an expression.

Parameters: variables in the procedure/function definition that receive the values (arguments) passed in when it's called. (IGCSE limits this to a maximum of 2 parameters.)

```
PROCEDURE Stars (Number : INTEGER)
  FOR Counter ← 1 TO Number
    OUTPUT "*"
  NEXT Counter
ENDPROCEDURE

CALL Stars(7)

FUNCTION Celsius (Temperature : REAL) RETURNS REAL
  RETURN (Temperature - 32) / 1.8
ENDFUNCTION

MyTemp ← Celsius(98.6)
```

Local vs global variables

Global variable: scope = the WHOLE program — can be used anywhere.

Local variable: scope = only the procedure/function it was declared inside — using it outside gives an error ("undefined").

Common mistake: if a variable is declared BOTH globally and locally with the same name, changes made to the local copy inside the procedure do NOT affect the global copy outside it.

8.1.7 Library Routines

<i>Routine</i>	<i>Purpose</i>	<i>Example</i>
MOD	remainder of division	MOD(10,3) = 1
DIV	whole-number quotient	DIV(10,3) = 3
ROUND	round to N decimal places	ROUND(6.97354,2) = 6.97
RANDOM	random number (0 to 1 inclusive)	RANDOM()

8.1.8 Creating a Maintainable Program

A maintainable program should:

- use meaningful identifier names for all variables, constants, arrays, procedures & functions
- be divided into modules (procedures/functions) for each task
- be fully commented (Python `#`, VB `'`, Java `//` or `/* */`)

8.2 Arrays

Array: a data structure holding several elements of the SAME data type under one identifier; each element is accessed by its index (position number).

First index is usually 0. Declaring in pseudocode states name, index range(s) and data type:

```
DECLARE MyList : ARRAY[0:9] OF INTEGER           // one-dimensional (a
"list")
DECLARE MyTable : ARRAY[0:9,0:2] OF INTEGER       // two-dimensional (a
"table": rows,columns)

MyList[4] ← 27                                   // set one element
FOR Counter ← 0 TO 9
    INPUT MyList[Counter]                       // populate with a loop
NEXT Counter

FOR Row ← 0 TO 9                                // 2D array needs NESTED loops
    FOR Col ← 0 TO 2
        INPUT MyTable[Row,Col]
    NEXT Col
NEXT Row
```

Note: Python doesn't have true arrays – it uses lists (which CAN mix data types) and lists-of-lists for 2D structures.

8.3 File Handling

Files store data permanently (RAM contents are lost when the computer is switched off). Standard sequence: OPEN → read/write → CLOSE.

```
OPEN MyFile FOR WRITE
    OUTPUT "Please enter a line of text"
    INPUT TextLine
    WRITEFILE, TextLine
CLOSEFILE(MyFile)

OPEN MyFile FOR READ
    READFILE, TextLine
```



```
OUTPUT TextLine  
CLOSEFILE(MyFile)
```

Exam tip: always CLOSE a file after use — this saves changes and frees the file for other programs to access.

Key Terms

Variable / Constant

changeable / fixed named data store

Sequence

order statements are executed

Selection

choosing a path (IF/CASE)

Iteration

repeating code (loops)

Procedure

named routine, no return value

Function

named routine, DOES return a value

Parameter

variable receiving a passed argument

Local/Global variable

restricted / whole-program scope

MOD / DIV

remainder / quotient of division

Array

indexed collection of same-type data

Index

position of an element in an array

File

permanent stored collection of data

Practice Questions

- Q1.** Explain the difference between a procedure and a function. [3]
- Q2.** State the values of MOD(23,5), DIV(23,5), ROUND(3.14159,2) and 23 MOD 5 written using Python's operator. [4]
- Q3.** Write pseudocode for a function `Square` that takes one integer parameter and RETURNS its square. Then show how it would be called to store the square of 9 in a variable `Result`. [6]
- Q4.** A one-dimensional array `Scores` holds 15 integers (index 0 to 14). Write pseudocode using a FOR loop to input all 15 values from the keyboard. [4]
- Q5.** Explain what is meant by the scope of a variable. Using an example, explain the difference between a local and a global variable. [5]
- Q6.** Explain why a program should always CLOSE a file after writing data to it. [3]
- Q7.** Identify and correct the errors in this pseudocode, which should output the numbers 1 to 5: [5]

```
FOR Counter ← 1 TO 5  
    OUTPUT Count  
NEXT Count
```

Mark Scheme

- A1.** A procedure is a named block of code that performs a task but does NOT return a value (1); a function is a named block of code that ALWAYS returns a value (1), using RETURN, and can be used directly in an expression (1).
- A2.** MOD(23,5) = 3 (1); DIV(23,5) = 4 (1); ROUND(3.14159,2) = 3.14 (1); Python: 23 % 5 (1).

A3. Correct header with one INTEGER parameter and RETURNS INTEGER (1); RETURN statement calculating Number*Number or Number² (1); ENDFUNCTION (1); correct call e.g. Result ← Square(9) (1); assignment to Result correctly shown (1); appropriate variable naming (1).

A4. FOR loop from 0 to 14 (1 for correct bounds); correct loop variable used as index (1); INPUT Scores[Counter] inside loop (1); NEXT Counter correctly placed (1).

A5. Scope = the part of the program in which a variable can be accessed/used (1). Global: can be used/accessed by any part of the whole program (1). Local: can only be used within the procedure/function it is declared in (1). Example showing a variable declared inside a procedure being inaccessible outside it (2).

A6. Any three: ensures data is properly saved/written to the file (1); releases the file so other programs/parts of the program can access it (1); prevents data loss/corruption (1); frees up system resources (1).

A7. Errors: OUTPUT statement uses "Count" instead of "Counter" (1); NEXT statement uses "Count" instead of "Counter" (1). Corrected code: FOR Counter ← 1 TO 5 / OUTPUT Counter / NEXT Counter (3 marks for fully correct rewrite).

– end of Chapter 8 –